



DESIGN OF MECHATRONICS 1 - QUIZ

Sec 2A - Team 1

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Design Of Mechatronics 1 – Work Breakdown Structure

1. Planning

1.1 Design Sketches

1.1.1 Decide on which mechanisms to use.

1.1.2 Design sketches for each subsystem.

1.1.3 Sketch the integrated subsystems.

1.2 WBS

1.2.1 Work Breakdown Structure generation for all project milestones.

1.3 Gantt Chart

1.3.1 Plan tasks based on submission times and identify critical tasks.

1.4 Bill Of Materials

1.4.1 Identify electronics/mechanical parts and their prices.

2. Design

2.1 CAD Design

2.1.1 3D Model the first subsystem. (Bi-level sorting)

2.1.2 3D Model the second subsystem. (3D Cartesian Robot)

2.2 Manufacturing Plan

2.2.1 Identify the most suitable manufacturing technique/s for each part.

2.2.2 Minimize the use of 3D printed parts & prioritize standard parts.

2.3 Actuator Sizing

2.3.1 Identify the actuators in each subsystem.

2.3.2 Perform the suitable calculations based on the loads each actuator is exposed to.

2.4 Simulation

2.4.1 Import CAD designs into MATLAB.

2.4.2 Use the toolboxes in MATLAB to run simulations of each subsystem and of the system.

2.4.3 Perform PID control simulations on one of the motors and tune the PID values.

2.4.4 Create a StateFlow model that describes the system.

3. Subsystems

3.1 Stress Analysis

3.1.1 Perform FEA on critical/load bearing parts of each subsystem.

3.2 Manufacturing / Assembly

3.2.1 Manufacture each part using different manufacturing techniques.

3.2.2 Assemble each part correctly as a part of the system/subsystem.

3.3 Programming / Electrical Part 1

3.3.1 Programming the microcontrollers for the basic functionality of each subsystem.

3.3.2 Integrate logical and power circuits without damage.

3.4 Programming / Electrical Part 2

3.4.1 Implement FreeRTOS.

3.4.2 Establish communication between microcontrollers through ESP-NOW.

5. Final Implementation

5.1 Integration

- 5.1.1 Align the output of the first subsystem with the input of the second.
- 5.1.2 Establish microcontroller communication between the subsystems.
- 5.1.3 Wire management/Control box implementation.

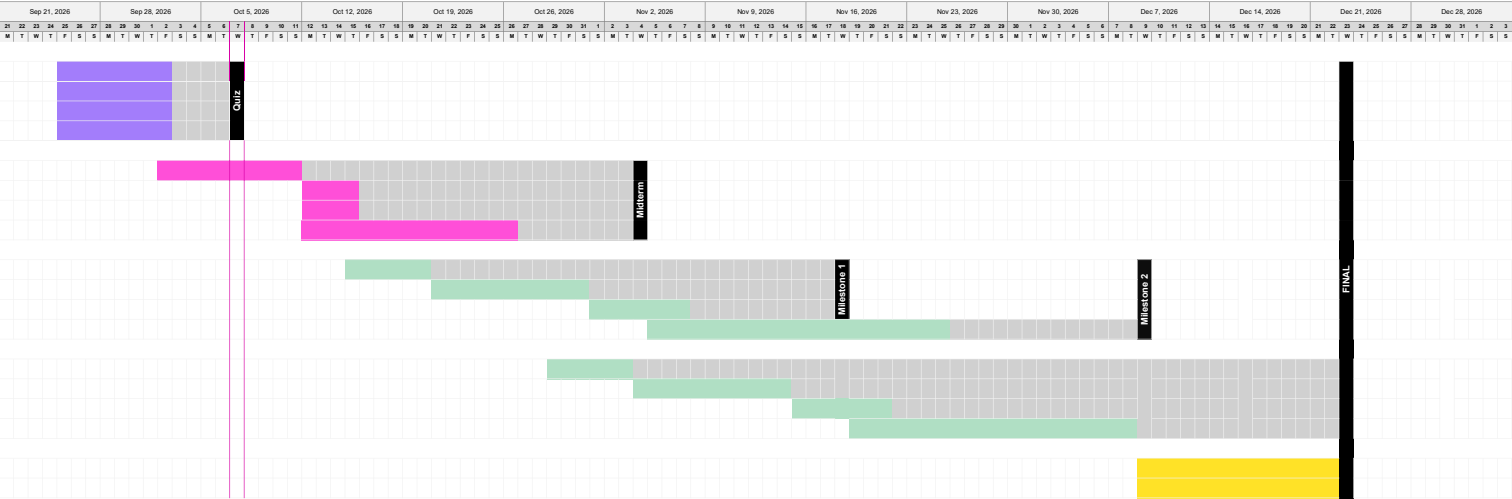
5.2 Testing

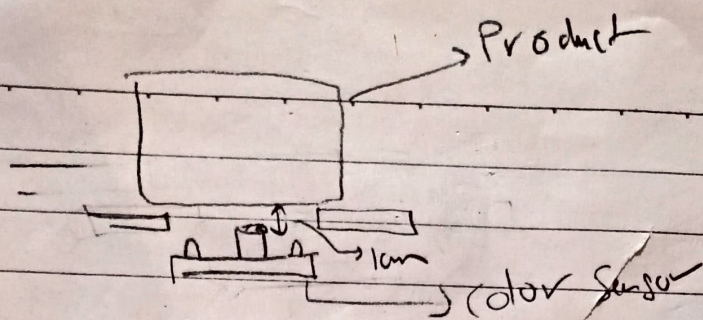
- 5.2.1 Run the full system, identify issues, resolve any that come up.

Design Of Mechatronics 1

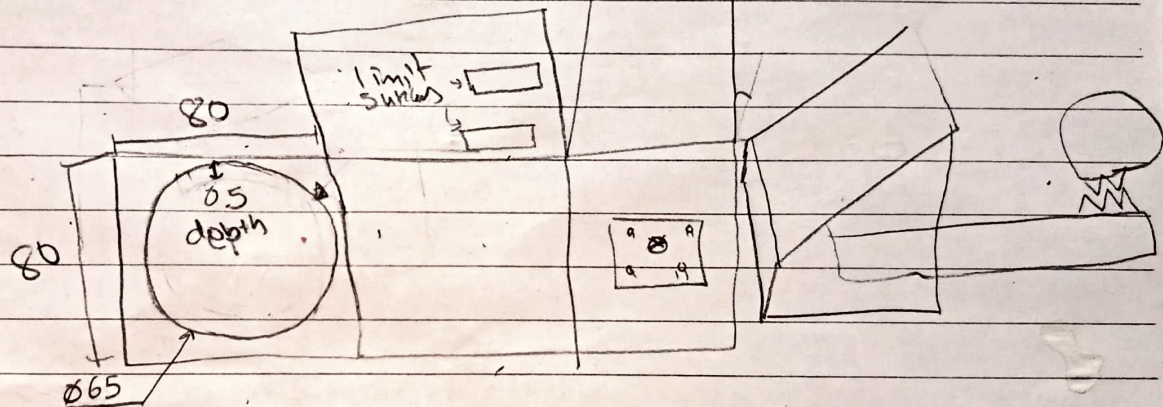
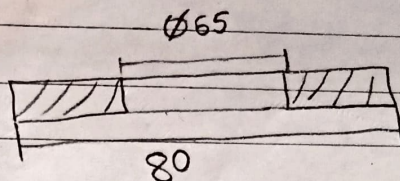
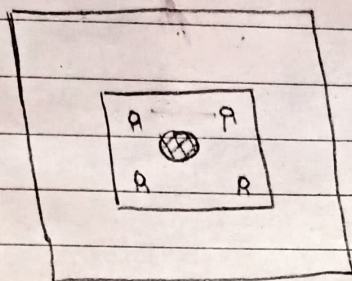
Project for Aly				
Team Members:	Youssef	Self	Kerolos	Ataf
TASK	ASSIGNED TO		START	END
Initiation				
Design Sketches	Team		2026-09-25	2026-10-02
WBS	Aly		2026-09-25	2026-10-02
Chartt Chart	Aly		2026-09-25	2026-10-02
BOM Materials	Self		2026-09-25	2026-10-02
Planning and design				
CAD Design	Youssef + Self		2026-10-02	2026-10-11
Manufacturing Plan	Ataf		2026-10-12	2026-10-15
Actuator Sding	Aly + Ataf		2026-10-12	2026-10-15
Simulation	Self + Kerolos		2026-10-12	2026-10-26
Subsystem 1				
Stress Analysis	Youssef		2026-10-15	2026-10-20
Manufacturing/Assembly	Ataf		2026-10-21	2026-10-31
Programming/ Electrical Part 1	Aly + Kerolos		2026-11-01	2026-11-07
Programming/ Electrical Part 2	Aly+ Ataf		2026-11-05	2026-11-25
Subsystem 2				
Stress Analysis	Self		2026-10-29	2026-11-03
Manufacturing/Assembly	Youssef + Kerolos		2026-11-04	2026-11-14
Programming/ Electrical Part 1	Youssef + Ataf		2026-11-15	2026-11-21
Programming/ Electrical Part 2	Aly+ Self		2026-11-19	2026-12-09
Evaluation				
Integration	Team		2026-12-09	2026-12-22
Testing	Team		2026-12-09	2026-12-22

Project start: Fri, 9/25/2026
Display week: 1

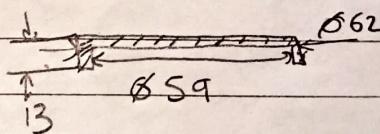
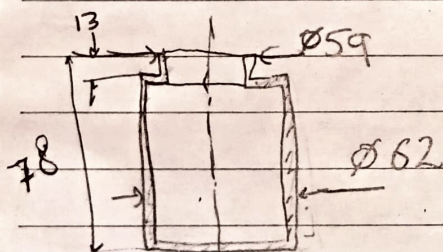




first spot



Product

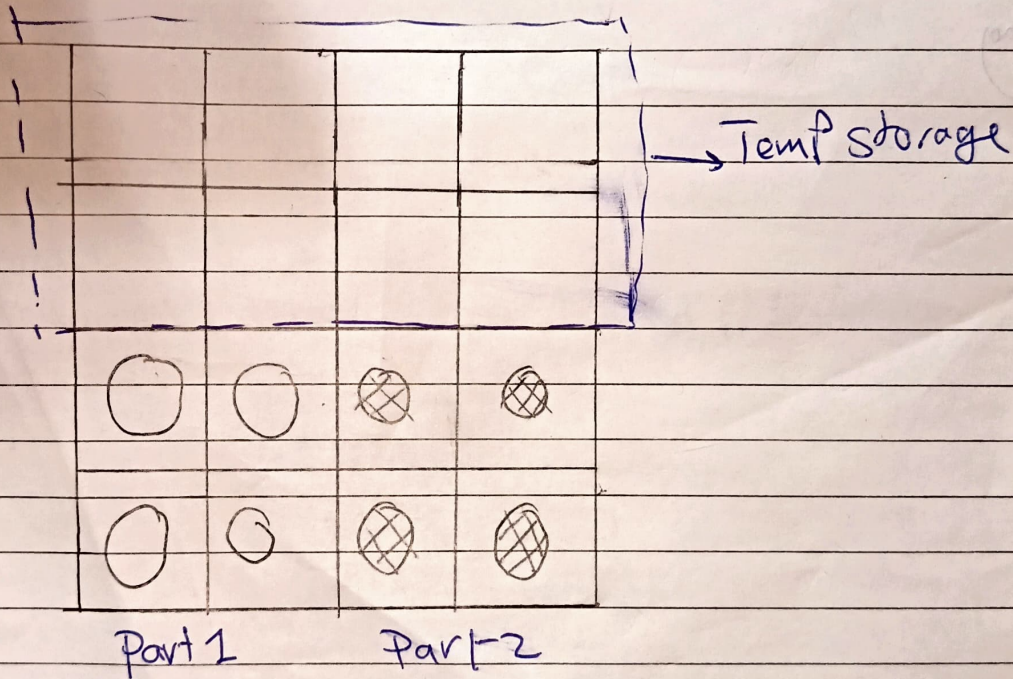
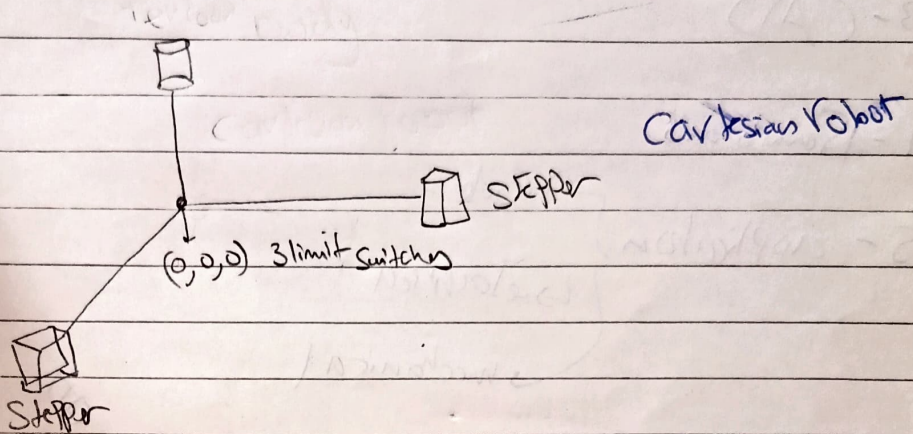
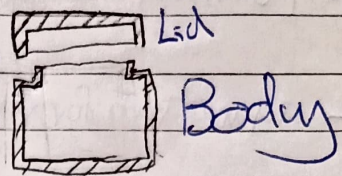


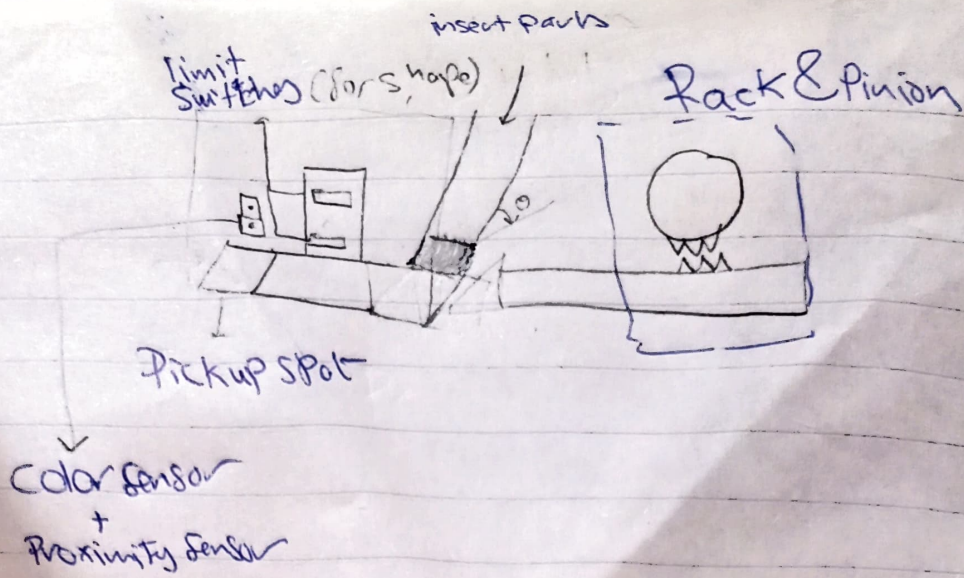
Body

lid

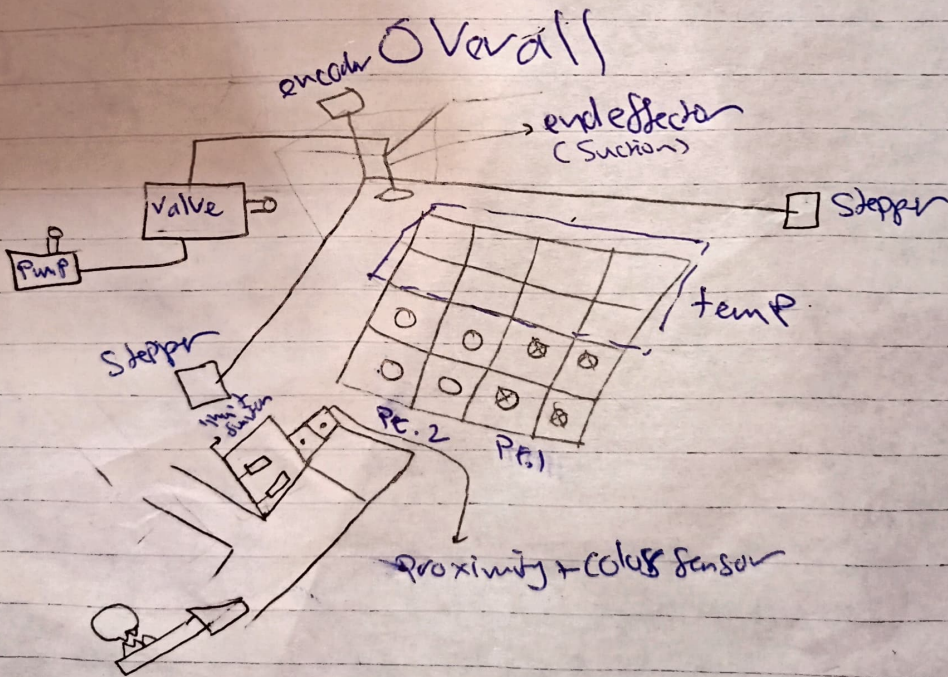
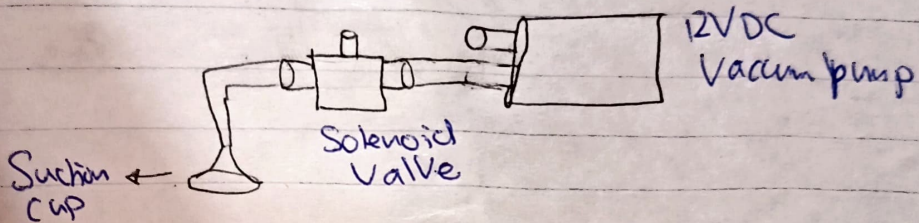
System 1:

Product:





End effector:



BILL OF MATERIALS			
Part	Quantity	Price	Notes
Limit switch	5	40	
ESP32	2	560	(-280 for reuse from Basics of Mechatronics)
Diode	5	20	(-20 for reuse from Basics of Mechatronics)
Cooling Fan	1	50	
Color Sensor	1	325	
Vaccum Pump	1	350	Suction system to hold the parts
Stepper Motor	2	1000	High load carrying capacity + easy to identify position
Stepper Driver	2	200	
Timing belt system	2	1400	Cheap & Reliable
Motor + Encoder	1	500	For closed loop PID
Power Screw	1	1000	Easy to maintain
Laser Printing	1	2000	Cheap, fast , and easy manufacturing method
3D printing	1	1500	for non standard parts
Misc	1	2000	Buffer
Sum		10945	EGP